Supplementary Information

Probabilistic graph completion of historical Chinese incense formulas under source shift

Supplementary Note 1 Data curation decisions

The reference target is the relative dry-mass composition of aromatic materials in one burned or fumigating formula. Honey, fruit juice, wine, charcoal and binders remain in the source record as process materials but do not enter the aromatic composition vector. Curation retained only formulas whose aromatic quantities could be converted without imputation. Every record contains the original material string, canonical candidate, identity status, historical quantity string, normalized qian value and source locator.

The curation manifest records 91 inclusions and 29 exclusions. Exclusion categories include missing aromatic quantities, incomplete extraction boundaries, prose without an executable formula and unresolved duplication. The family audit records title similarity, material overlap, ratio-pattern evidence and source genealogy for all retained candidates.

Supplementary Note 2 Split construction

The development set contains 85 formulas from five work groups. Five leave-one-work-out folds supported hyperparameter selection. Ten primary folds held out each unordered pair of work groups. Six Hong-source formulas remained outside vocabulary construction, feature statistics, model fitting and primary evaluation until the confirmation run. Researchers had already handled these records during curation. The confirmation is model-quarantined; it is not investigator-blinded.

Supplementary Note 3 Leakage controls

For every training formula, global prevalence, source prevalence, title-material centroids and the directed material graph excluded that formula. Hyperparameters were selected only from development leave-one-work-out results. The primary source-pair metrics did not change model parameters. The Hong confirmation ran after code and model configuration were frozen.

Supplementary Note 4 Model selection

The grid crossed five Gaussian-prior variances, two training objectives and three graph reducers over five mask seeds. The selected full decoder used $C = 3.0$, empirical-risk minimization and arithmetic-mean graph affinity. Development source-macro hidden F1 was 0.3911 with standard deviation 0.0081 across seeds. The strongest development baseline, graph-only with $C = 0.1$ and eight group-DRO updates, reached 0.3876 with standard deviation 0.0116. The mean paired seed difference was 0.0035, with descriptive 95% t interval $[-0.0040, 0.0110]$.

Supplementary Note 5 Approximate Bayesian inference checks

The inclusion model uses a Gaussian coefficient prior and a Laplace posterior centered at the maximum a posteriori solution. A separate Metropolis-Hastings chain with 20,000 draws checked posterior predictive probabilities on a synthetic logistic problem. The mean absolute difference between Laplace and Monte Carlo predictive probabilities was 0.0016.

Synthetic structured formulas provided an implementation-level recovery test. The proposed decoder reached hidden-component F1 0.900 and weighted Jaccard 0.658; frequency reached 0.497 and 0.445, respectively. This test verifies that the code can recover signal generated with compatible structure. It is not part of the historical performance estimate.

Supplementary Note 6 Chemical representations

The material evidence graph contains 18 profile records covering 18 ontology materials, 17 analytical publications and 40 PubChem compounds. Each profile retains specimen, taxon, plant part, preparation, analytical method and measurement basis. Presence-only compounds receive equal conditional weights within that profile and are never described as measured abundance.

The material kernel combines three views: weighted Morgan fingerprint bits, standardized physicochemical descriptors and controlled odor tags. Similarity weights are 0.50, 0.25 and 0.25. The main paper reports the negative development and auxiliary-task results because they determine the model's scientific interpretation.

Supplementary Note 7 Metric definitions

For true hidden set H and predicted hidden set P,

precision = $|H \cap P| / |P|$,

recall = $|H \cap P| / |H|$,

F1 = 2 precision recall / (precision + recall).

For reference and predicted composition vectors $\mathbf{w}$ and $\hat{w}$, weighted Jaccard is

$\Sigma_j \min(w_j, \hat{w}_j) / \Sigma_j \max(w_j, \hat{w}_j)$.

Expected calibration error uses ten equal-width probability bins and weights each absolute probability-frequency gap by bin count. The Brier score averages squared error over hidden in-vocabulary candidate decisions.

Supplementary Note 8 Reproducibility commands

From the research directory:

```
uv sync --all-groups
make cycle2-data
uv run python -m xiangji.freeze_cycle3
uv run python -m xiangji.freeze_cycle4
make cycle5-manuscript
```

```
make verify-references
make export-submission
make export-latex
make clean-reproduction
make cycle6-release
```

The release manifests bind source data, configuration, experiment and report files by SHA-256. Raw primary and confirmation records are stored as deterministic gzip archives with modification time fixed to zero.

Supplementary Note 9 Competitive baselines and post-freeze analyses

Formula k-nearest neighbours and PureSVD were added after the primary decoder was frozen. Their configuration grids used only the five development work splits. Formula k-nearest neighbours selected ten cosine neighbours; PureSVD selected rank two. The unchanged primary folds then provided their first source-pair results.

The cross-work similarity analysis flagged nine formulas whose aromatic material sets reached Jaccard 0.8 against another work group. Removing them lowered every model's score but preserved the graph-frequency ordering. The released table lists all reference pairs and similarities.

The original 80% count interval covered 72.6% of the 420 work-held-out development predictions. Integer distance from the interval supplied the nonconformity score. The finite-sample 80% quantile was one material, which expanded the primary interval from 64.2% to 82.5% coverage. Its mean width increased from 1.24 to 2.86 materials.

Supplementary Note 10 Case-level reconstruction

The case rule selected the seed-17 Hong formula closest to the median hidden F1 across that run. It selected *Yinxiang fa*, with six observed and five hidden reference materials. Three hidden materials were recovered, two were missed and two extra materials were selected. Supplementary Table S8 gives the complete input, prediction and reference composition. The output holds observed ratios fixed and allocates the remaining mass across predicted hidden materials.

Supplementary Tables

Supplementary Table S1

work_id	work	formula_count	unique_materials	median_formula_size	role
chen_xiangpu_compilation	Chen Xiangpu	38	50	6.5	development_and_primary
hong_xiangpu_attributed	Hong Xiangpu	6	18	9.0	confirmation
huizhai_xiangpu	Huizhai Xiangpu	10	27	4.5	development_and_primary
liexiang_xinpu	Liexiang Xinpu	6	32	9.0	development_and_primary
moe_xiaolu_xiangpu	Xiaolu Xiangpu	2	10	7.0	development_and_primary
xiangcheng_compilation_unresolved_source	Xiangcheng compilation	29	42	6.0	development_and_primary

Supplementary Table S2

model	hidden_f1_source_macro	weighted_jaccard_source_macro	aitchison_distance_source_macro	development_formula_count	held_out_work_groups	mask_seeds
Frequency	0.3386314231241636	0.37880861819102574	7.890524747021504	85	5	5
Directed graph	0.36541975977910646	0.3879966344368349	7.716927225587732	85	5	5
Full decoder	0.3664678698662365	0.3871368821278717	7.723978147672447	85	5	5

Supplementary Table S3

work_id	work	model	record_count	hidden_f1	weighted_jaccard	aitchison_distance
chen_xiangpu_compilation	Chen Xiangpu	Frequency	38	0.3558255487202856	0.37332281842456044	7.835823982693473
chen_xiangpu_compilation	Chen Xiangpu	Directed graph	38	0.40381787802840435	0.3859633269601026	7.5774945991539475
chen_xiangpu_compilation	Chen Xiangpu	Full decoder	38	0.3959137426900585	0.38210883346643226	7.628889444433014
huizhai_xiangpu	Huizhai Xiangpu	Frequency	9	0.1776388888888889	0.3111130478356962	9.080651238953031
huizhai_xiangpu	Huizhai Xiangpu	Directed graph	9	0.21828703703703703	0.32335090014279483	8.85998210060979
huizhai_xiangpu	Huizhai Xiangpu	Full decoder	9	0.20981481481481482	0.32409093041018855	8.922589444436754
liexiang_xinpu	Liexiang Xinpu	Frequency	6	0.42838564213564206	0.3677523014100504	6.757115913052538
liexiang_xinpu	Liexiang Xinpu	Directed graph	6	0.44623075998075995	0.37521044493583994	6.583982562099366
liexiang_xinpu	Liexiang Xinpu	Full decoder	6	0.4430489417989418	0.37729351929577887	6.599756898677163
moe_xiaolu_xiangpu	Xiaolu Xiangpu	Frequency	2	0.39999999999999997	0.4513607447979179	7.815560804825541
moe_xiaolu_xiangpu	Xiaolu Xiangpu	Directed graph	2	0.375	0.4514001317458871	7.893871154345915
moe_xiaolu_xiangpu	Xiaolu Xiangpu	Full decoder	2	0.3916666666666666	0.4458173084000894	7.865445215265594
xiangcheng_compilation_unresolved_source	Xiangcheng compilation	Frequency	29	0.33130703587600147	0.3904941784869038	7.96347179558294
xiangcheng_compilation_unresolved_source	Xiangcheng compilation	Directed graph	29	0.38376312384933076	0.40405836839955	7.669305711729642
xiangcheng_compilation_unresolved_source	Xiangcheng compilation	Full decoder	29	0.3918951833607006	0.4063738190668692	7.6032097355497115

Supplementary Table S4

observed_fraction	model	full_precision	full_recall	full_f1	hidden_precision	hidden_recall	hidden_f1	count_absolute_error	weighted_jaccard	union_ratio_mae	aitchison_distance	inclusion_brier_in_vocabulary	true_material_vocabulary_coverage
0.25	Frequency	0.5770524089077781	0.5205203538606469	0.535165920499632	0.4338726966571563	0.37662478021127344	0.38822547355791 23	1.5986882113482097	0.2692787727935844	0.1419400548 7909123	8.7608163316 95722	0.0575087567 67470616	0.8780815174 89564
0.25	Directed graph	0.5928240456650455	0.5347500029748115	0.5499278971881092	0.4550929143 8663524	0.3952915067 2504343	0.4078295186 5053685	1.5986882113482097	0.2815179959 5651417	0.1400361012 977414	8.6192421001 7007	0.0565176971 3786037	0.8780815174 89564
0.25	Full decoder	0.5859556399485062	0.5290100743167427	0.5437097062 6252	0.4457206285 1786953	0.3876623399 2855537	0.3995121705 7927413	1.5986882113482097	0.2799527118122218	0.1396667671 7124277	8.6682894137 389	0.0572109470 3027725	0.8780815174 89564
0.75	Frequency	0.8009657009853495	0.7060492692868279	0.7423549434359937	0.3036700476 6966016	0.2193554172 0014848	0.2372878577 289895	1.1279674981645897	0.5348493099051181	0.1030142207 4909235	6.6202649392 78646	0.0234350882 49582618	0.8780815174 89564
0.75	Directed graph	0.8086106349951695	0.7134189016101731	0.7498408911223192	0.3318699665102005	0.2460351834590448	0.2620843337 3812134	1.1279674981645897	0.5405423250690568	0.1025646870 3498842	6.4775733175 91468	0.0256371777 5493035	0.8780815174 89564
0.75	Full decoder	0.8088870602064324	0.7136102396 73875	0.7500697530333453	0.3326837721467529	0.2461639868 9972546	0.2626371954 8764286	1.1279674981645897	0.5362687243345816	0.1032364714 2929156	6.4912674319 9254	0.0274388472 72256688	0.8780815174 89564
0.5	Frequency	0.6925325671075753	0.6077382872283252	0.6363095091504576	0.3998593544485255	0.3193591468 9730227	0.3379089035 5867537	1.3256411824685241	0.3779591583891666	0.1293384525 904721	7.8736924324 08438	0.0402178460 1719576	0.8780815174 89564
0.5	Directed graph	0.7146706893592614	0.6285879170764608	0.6576549815333873	0.4433774601 03154	0.3592957409 508147	0.3787410917 430981	1.3256411824685241	0.3897618288188972	0.1284187240 2246954	7.6363587279 74793	0.0391413262 14132556	0.8780815174 89564
0.5	Full decoder	0.7126867149963019	0.6282488294523613	0.6564365904 29974	0.4388407151 109023	0.3615531566 366319	0.3768267595 605529	1.3256411824685241	0.3897642797262614	0.1279232309 9579855	7.6461730350 75958	0.0402186673 858507	0.8780815174 89564

Supplementary Table S5

frequency_class	recovered	total	recall
common_5_plus	2221	3876	0.5730134158926729
out_of_vocabulary	0	1445	0.0
rare_2_to_4	44	787	0.05590851334180432
singleton	8	342	0.023391812865497075

Supplementary Table S6

lower	upper	n	mean_probability	empirical_frequency
0.0	0.1	82012	0.03655585734914713	0.021252987367702286
0.1	0.2	6100	0.1373533179318377	0.1198360655737705
0.2	0.3	1958	0.2441597979803616	0.23084780388151174
0.3	0.4	1153	0.3473377310162553	0.30355594102341715
0.4	0.5	943	0.4484651791966857	0.35949098621421
0.5	0.6	900	0.5507251591130595	0.35333333333333333
0.6	0.7	767	0.6484393439988508	0.45241199478487615
0.7	0.8	639	0.74895977597897	0.4945226917057903
0.8	0.9	531	0.8485609517922754	0.5499058380414312
0.9	1.0	207	0.931495340440959	0.5652173913043478

Supplementary Table S7

model	full_precision	full_recall	full_f1	hidden_precision	hidden_recall	hidden_f1	count_absolute_error	weighted_jaccard	union_ratio_mae	aitchison_distance	inclusion_brier_in_vocabulary	true_material_vocabulary_coverage
Frequency	0.7565656565656566	0.7065656565656565	0.7287878787878789	0.5022222222222222	0.4466666666666666	0.4688888888888889	0.5	0.4195449795313896	0.0899240874086868	7.270551718135342	0.039167612958608065	0.9833333333333334
Directed graph	0.8045454545454547	0.7503787878787879	0.7744528619528619	0.5999999999999999	0.5305555555555557	0.5583333333333333	0.5	0.4560198542263435	0.0863877911868412	6.674412262317662	0.033198851906774 35	0.9833333333333334
Full decoder	0.8170454545454545	0.7595454545454546	0.7851010101010101	0.625	0.5444444444444445	0.5766666666666667	0.5	0.4676302262881 4505	0.0849672670887 4247	6.530307597353823	0.032080442697 17458	0.9833333333333334

Supplementary Table S8

material_id	outcome	reference_ratio	predicted_ratio	inclusion_probability
agarwood	observed input	0.06648936170212766	0.06648936170212766	1.0
agarwood_zhan_grade	recovered hidden	0.3324468085106383	0.1170807281430485	0.1579713133268401
borneol_family	recovered hidden	0.013297872340425532	0.09916713200242623	0.6169273386166794
clove_bud	false positive	0.0	0.1059280314005388	0.2841450773200888
frankincense	missed hidden	0.06648936170212766		
licorice	missed hidden	0.06648936170212766		
linglingxiang	false positive	0.0	0.11995743581414034	0.44027671710104416
mao_aromatic_grass	observed input	0.26595744680851063	0.26595744680851063	1.0
musk	observed input	0.05319148936170213	0.05319148936170213	1.0
onycha	observed input	0.0013297872340425532	0.0013297872340425532	1.0
patchouli	observed input	0.0013297872340425532	0.0013297872340425532	1.0
sandalwood_white	recovered hidden	0.06648936170212766	0.103079438597293	0.6953282835587735
spikenard	observed input	0.06648936170212766	0.06648936170212766	1.0

Supplementary Table S9

work_id	formula_count	source_pages
chen_xiangpu_compilation	38	Chen shi xiangpu (Siku Quanshu edition)/juan 2; Chen shi xiangpu (Siku Quanshu edition)/juan 3
hong_xiangpu_attributed	6	Xiangpu (Siku Quanshu edition)/juan xia
huizhai_xiangpu	10	Xiangcheng (Siku Quanshu edition)/juan 23
liexiang_xinpu	6	Xiangcheng (Siku Quanshu edition)/juan 25
moe_xiaolu_xiangpu	2	Xiangcheng (Siku Quanshu edition)/juan 24
xiangcheng_compilation_unresolved_source	29	Xiangcheng (Siku Quanshu edition)/juan 14; Xiangcheng (Siku Quanshu edition)/juan 15; Xiangcheng (Siku Quanshu edition)/juan 16; Xiangcheng (Siku Quanshu edition)/juan 17; Xiangcheng (Siku Quanshu edition)/juan 18; Xiangcheng (Siku Quanshu edition)/juan 21

Supplementary Table S10

model	hidden_f1	weighted_jaccard
Frequency	0.3386314231241636	0.37880861819102574
Formula kNN	0.34311186392924137	0.4434976232018514
PureSVD	0.33434939272598074	0.4199705870605026
Directed graph	0.36541975977910646	0.3879966344368349
Full decoder	0.3664678698662365	0.3871368821278717

Supplementary Table S11

model	all_formulas_hidden_f1	after_excluding_high_similarity_hidden_f1	excluded_formula_count	jaccard_threshold
Frequency	0.3386314231241636	0.32233192269010447	9	0.8
Directed graph	0.36541975977910646	0.3481871861471862	9	0.8
Full decoder	0.3664678698662365	0.34937990860990864	9	0.8

Supplementary Table S12

interval	coverage	mean_width
Base posterior	0.641566265060241	1.2439759036144578
Development-calibrated	0.8253012048192772	2.86144578313253